Language Identification from Audio

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Abstract

This project explores spoken language identification (SLID) under limited speaker diversity, where models have a likelihood of learning speaker specific acoustic patterns rather than language level characteristics. A baseline system using MMS-300m with 31.96% validation accuracy was achieved and 35.67% with hyperparameter tuning. Pivoting to multil ingual mHuBERT-147 model and freezing the CNN feature encoder further significantly improved performance to 59.3%. To address speaker bias, dynamic audio augmentation was applied during training, including speed perturbation, pitch shifting, and Gaussian noise, which prevented shortcut learning through speaker specific cues. Confusion matrix and t-SNE analysis further confirmed that learned representations align more closely with language boundaries rather than speaker identity. Our results suggest that speaker bias in this setting is rather concerning which only augmentation cannot completely resolve, and may require more sophisticated approaches such as adversarial training via Domain Adversarial Neural Networks (DANN) or speaker embedding subtraction using pretrained speaker verification models.

1 Introduction

Spoken Language Identification involves recognizing the language directly from audio without depending on transcripts or text. Latest developments in self-supervised learning have driven models like wave2vec 2.0 (Baevski et al., 2020) and HuBERT (Hsu et al., 2021) to the forefront of speech processing, as they have been shown to capture rich acoustic features from large unlabeled speech. However, a less explored challenge arises when training data includes very few speakers per language. In such a case, model can learn to recognize voices rather than languages.This is a form of shortcut learning (Geirhos et al., 2020) and it quietly undermines how well the model generalizes to new speakers. In this work, the dataset contains five speakers per 22 official Indian languages, making this problem especially relevant. This study pursues three goals: establishing strong baseline system, mitigate speaker bias through data centric and regularization based strategies, and analyzing learned representations to verify whether model encodes language or speaker identity.

2 Methods

2.1 Model Architecture

SLID system is based on pretrained self-supervised speech encoders followed by classification layers. These models learn general acoustic patterns from large speech dataset and can be adapted to downstream tasks with limited labeled data.

Several pretrained encoders like mms-330m (Pratap et al., 2023), wave2vec and mHuBERT were explored. Among these , the multilingual mHuBERT-147 was selected (Boito et al., 2024) due to its ability to capture cross-lingual acoustic patterns that are instrumental for multilingual language identification.

2.2 Encoder Freezing Strategy

Early experiments showed unstable learning and validation accuracy close to random guessing, suggesting the model was overfitting to speaker-specific patterns rather than learning language features. During fine-tuning, the convolutional feature encoder was frozen to prevent overfitting to low-level speaker-specific patterns. Prior work has shown that lower layers of self-supervised speech models capture general acoustic features learned during large-scale pre-training (Pasad et al., 2021), while freezing lower layers during fine-tuning is an established strategy for preserving pretrained representations (Howard and Ruder, 2018). This allows the higher transformer layers to adapt to the language identification task without corrupting low-level acoustic features.

Model	Configuration	Val. Acc.
MMS-300m	Baseline, lr=1e-5, 5s	35.7%
Wav2Vec2	lr=1e-5, 5s	18.1%
mHuBERT-147	lr=1e-5, 5s	$\sim$9%
mHuBERT-147	lr=5e-6, 10 epochs	$\sim$12%
mHuBERT-147	Frozen encoder, lr=1e-5, 10s	59.3%
mHuBERT-147	Frozen encoder, 10s, no aug.	$\sim$62%
mHuBERT-147	**Frozen encoder, 10s, dynamic aug. + dropout**	**$\sim$61%**

Table 1: Validation accuracy across model configurations. Bold indicates the final submitted system.

2.3 Augmentation

To reduce speaker bias, we applied three audio augmentation methods: speed perturbation (Park et al., 2019), pitch shifting, and additive Gaussian noise (Seltzer et al., 2013).These transformations introduce variation in speech characteristics like fundamental frequency and speaking rate while also preserving linguistic content, which prevents the model from relying on speaker specific acoustic cues.

Two augmentation strategies were explored. Static augmentation was applied during preprocessing to get a fixed augmented version of each sample. Dynamic augmentation was applied during training time inside a custom `AugmentedAudioDataset` wrapper, generating a newly transformed version of each sample at every epoch (Cubuk et al., 2019). This increases acoustic diversity across epochs and more strongly discourages shortcut learning based on speaker identity. Further, a dropout regularization was applied to the transformer layers to reduce overfitting on the small training set.

3 Experimental Setup

Experiments were conducted on a speech dataset consisting of audio recordings for 22 official Indian languages with exactly five speakers per language and 400 samples per language in the training split. All audio was resampled to 16 kHz and truncated to a maximum duration of 10 seconds before feature extraction.

The models were trained using HuggingFace Transformers framework (Wolf et al., 2020) with AdamW optimizer, a learning rate of 1e-5, batch size of 16, and warmup ratio of 0.1 over 10 epochs. Weight decay of 0.01 was applied as additional regularization. The best checkpoint was selected based on validation accuracy using early stopping.

Several pretrained speech encoders and training strategies were evaluated. The most relevant configurations and their validation accuracies are summarized in Table 1. The final model uses the multilingual mHuBERT-147 encoder (Boito et al., 2024) with the convolutional feature extractor frozen during fine-tuning, which significantly improved performance over the MMS-300m baseline.

4 Results

4.1 Baseline and Model Improvement

The mms-300m baseline achieved 35.67% validation accuracy, however mHuBERT-147 significantly improved performance. When the feature encoder was frozen, validation accuracy increased to 59.3%. This suggests that the pretrained acoustic features were already well optimized and the full fine-tuning in a low-speaker setting could introduce overfitting.

4.2 Speaker Bias Mitigation Results

Static augmentation achieved 62% accuracy with a macro F1-score of 0.58. Dynamic augmentation combined with dropout achieved 61% accuracy with a similar macro F1-score. Although dynamic augmentation did not significantly increase overall accuracy, it still maintained stable performance under stronger regularization.

5 Analysis

To better understand model behavior, we analyzed the confusion matrices and t-SNE visualizations across four experimental conditions: the baseline (MMS-300m), mHuBERT-147 with frozen encoder and no augmentation, static augmentation, and dynamic augmentation.

Baseline vs. mHuBERT-147 with frozen encoder. The difference between the baseline and the frozen mHuBERT-147 model is obvious. The baseline confusion matrix (Appendix Figure 7) shows predictions scattered across many languages with very little structure along the diagonal, indicating the model struggles to separate most language pairs. The corresponding t-SNE (Appendix Figure 9) shows no clear language clusters, with points from

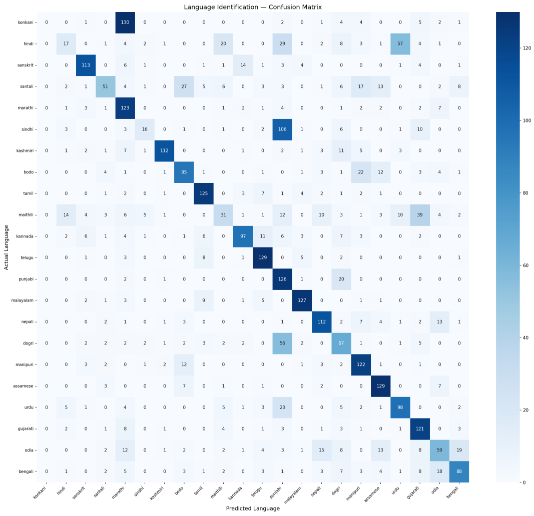

Figure 1: Confusion matrix (without augmentation)

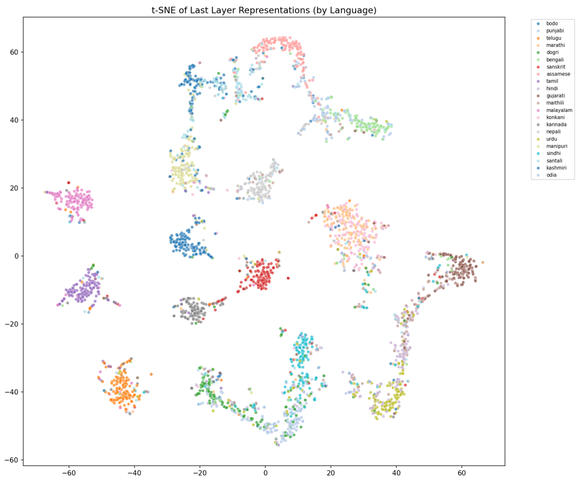

Figure 3: t-SNE visualization of embeddings-colored by language (without augmentation)

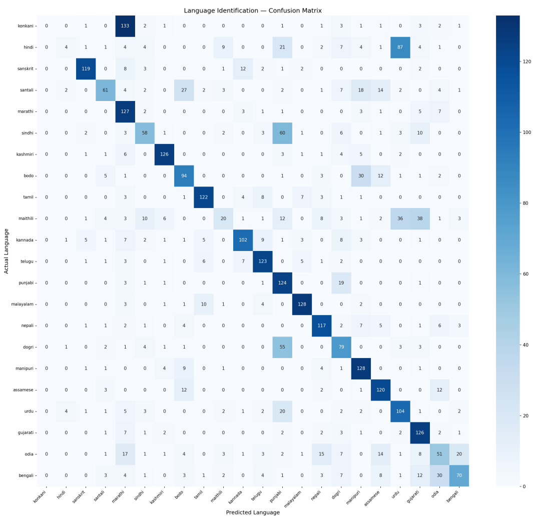

Figure 2: Confusion matrix (dynamic augmentation)

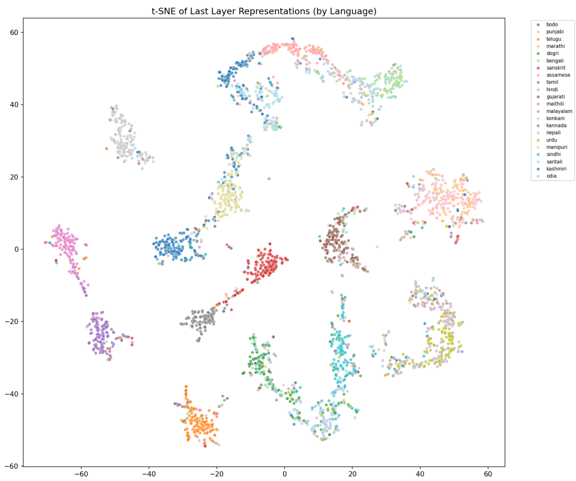

Figure 4: t-SNE visualization of embeddings- colored by language (dynamic augmentation)

different languages mixed together. In contrast, the no-augmentation mHuBERT-147 model (Figure 1) shows a much cleaner diagonal, and the t-SNE (Figure 3) shows visibly separated language clusters. This confirms that freezing the CNN encoder and switching to mHuBERT-147 was the most impactful change in our experiments.

Effect of dynamic augmentation. Although the no-augmentation and dynamic augmentation models achieve similar overall accuracy, the confusion matrices reveal differences in how individual languages are handled. Languages with distinct phonetic structures such as Tamil and Malayalam classified well by both models, with predictions concentrated along the diagonal in both Figure 1 and Figure 2.

Dynamic augmentation noticeably improved some of the more challenging languages. For example, correctly classified Santali samples increased from 51 to 61, and Kashmiri improved from 112 to 126. More notably, languages like Sindhi and Dogri showed meaningful gains with dynamic augmentation, suggesting these languages benefited from the increased acoustic diversity introduced during training. However, we also observed small drops in a few languages that were already well-separated. Bengali decreased from 88 to 70 correct predictions and Tamil from 125 to 122. This trade-off suggests that dynamic augmentation helps the model generalize better on harder language pairs at the cost of slightly reduced confidence on already-separable ones.

t-SNE analysis. The t-SNE visualizations provide further insight into this behavior. In the no-

Figure 5: t-SNE visualization of embeddings-colored by speaker (without augmentation)

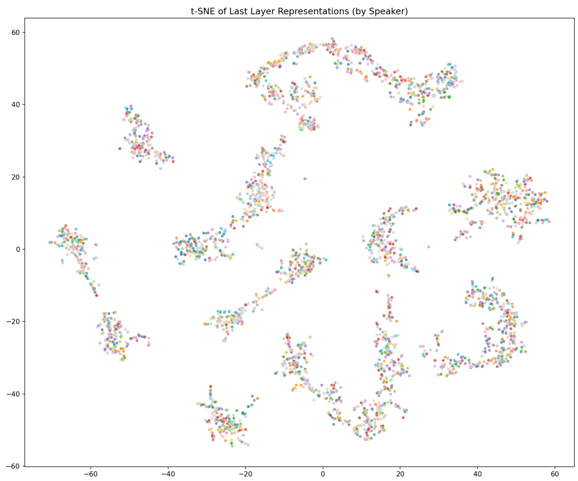

Figure 6: t-SNE visualization of embeddings-colored by speaker (dynamic augmentation)

augmentation model, language clusters are relatively compact and tightly grouped (Figure 3). In the dynamic augmentation model, clusters occupy a wider region of the embedding space (Figure 4), reflecting that the model was exposed to continuously varying acoustic patterns rather than fixed ones. Languages remain separable, but with greater internal variation within each cluster.

Speaker bias. In both models, the t-SNE colored by speaker identity (Figures 5 and 6) shows no clear clustering by speaker. Points from different speakers are intermixed within language regions rather than forming separate speaker groups. This is encouraging and suggests the model is primarily encoding language-level features rather than speaker identity. However, this result should be interpreted carefully. The absence of visible speaker clusters does not necessarily mean the model is free of speaker bias, since the validation set contains the same five speakers seen during training. It is possible that speaker information is encoded in a more subtle way that t-SNE does not reveal. The fact that augmentation improved performance on some languages while the speaker clustering remained similar across both conditions suggests the gains come from better language generalization rather than speaker bias reduction specifically. Fully resolving speaker bias would likely require more speakers in the dataset or more explicit methods such as adversarial training.

6 Conclusion

This study addressed spoken language identification under limited speaker diversity, where models risk learning speaker characteristics instead of language-level patterns. Switching to mHuBERT-147 with a frozen feature encoder considerably improved stability and generalization. Dynamic augmentation further encouraged speaker invariance by exposing the model to continuous acoustic variability during training, with representation analysis confirming that embeddings organize more strongly by language than by speaker identity.

7 Limitations

All experiments were conducted on the same dataset without an external validation set covering different speakers or acoustic conditions, limiting the scope of evaluation. Due to computational constraints, only a limited number of configurations could be explored, and further hyperparameter tuning may yield additional gains.

8 Future Work

Speaker bias remains partially unsolved. Adversarial training via DANN (Ganin et al., 2016) could force the model to learn speaker-invariant representations, while speaker embedding subtraction could remove speaker-specific information before classification. Ultimately, collecting data from more speakers per language would directly address the root cause of the problem.

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Appendix

This section includes the figures that are not included in the main report but are relevant. Here are the confusion matrix, tSNE for the static augmentation experiment and the baseline experiment.

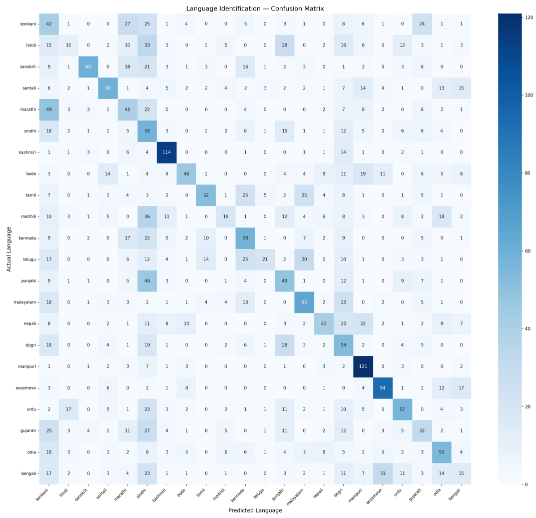

Figure 7: Confusion matrix (baseline)

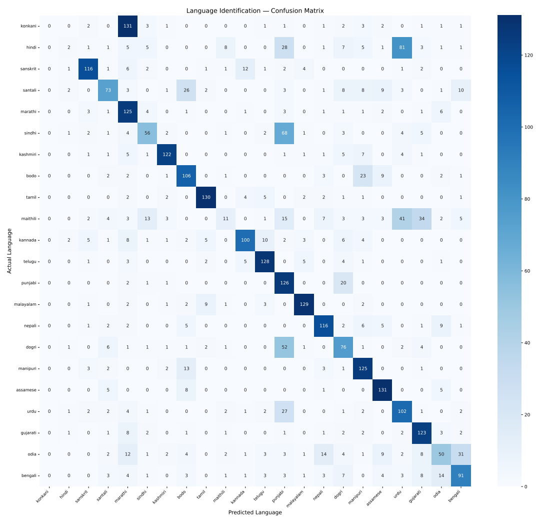

Figure 8: Confusion matrix (static augmentation)

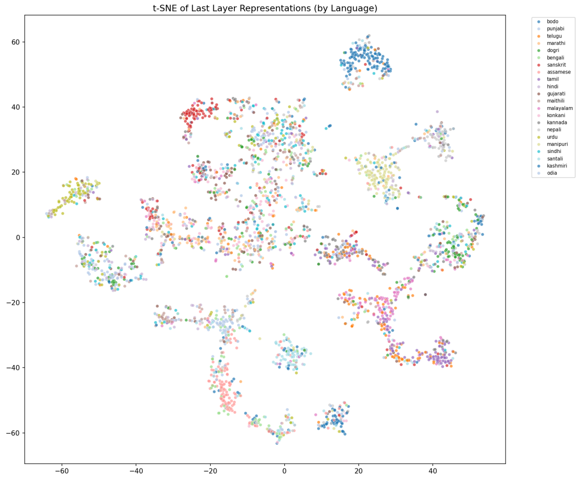

Figure 9: t-SNE visualization of embeddings - colored by language (baseline)

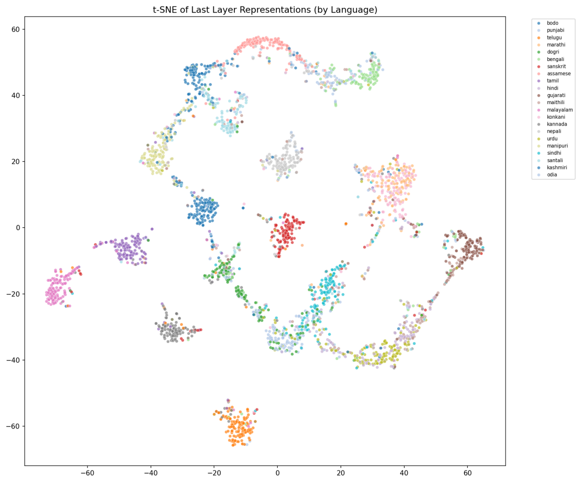

Figure 10: t-SNE visualization of embeddings - colored by language (static aug)

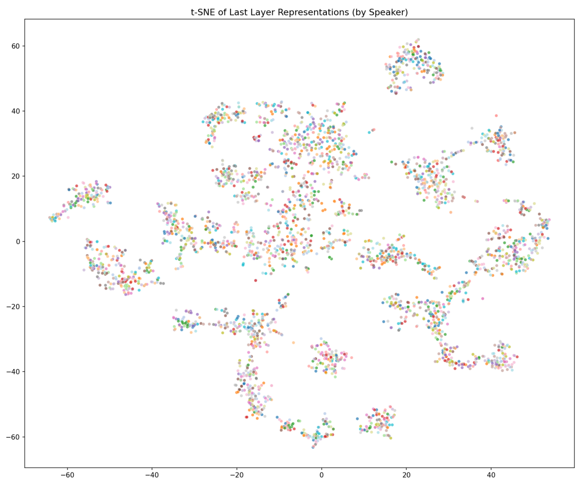

Figure 11: t-SNE visualization of embeddings - colored by speaker (baseline)

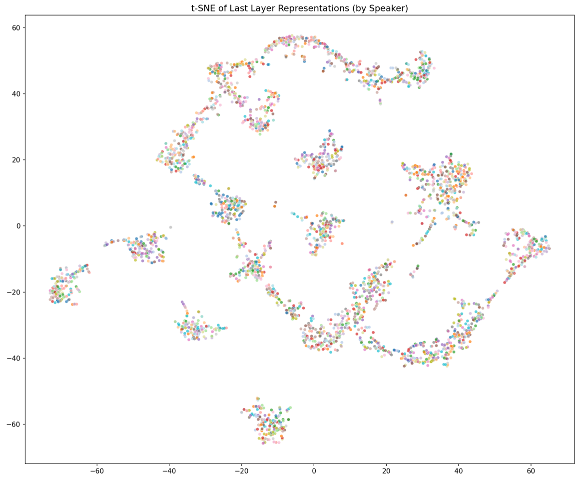

Figure 12: t-SNE visualization of embeddings - colored by speaker (static aug)